DISSERTATION PROGRESS REPORT & ONLINE APPENDIX

Chapter 2: Empirical Grounding, Capacity Reconstruction, and the Overaccumulation Regime (Specification A)

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Abstract

EXECUTIVE SUMMARY: This progress report provides the complete, paper-facing empirical appendix for **Specification A (Single Aggregate Capital Model)** of Chapter 2. I evaluate the full range of 168 estimated IM-OLS cointegrating regressions across four master comparative matrices comparing specifications **With Step Controls ($\mathbf{\Lambda D}'_t$ = Yes)** against **Without Step Controls ($\mathbf{\Lambda D}'_t$ = No)**: (1) **Full Sample Baseline Matrix (1929–2024)** across Centering Treatments; (2) **Accumulation Eras Comparative Matrix** comparing Fordism (1929–1973) vs. Post-Fordism (1974–2024); (3) **Multi-Wage Share Robustness Matrix** comparing Baseline NFC GVA, Corporate GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 wage measures; and (4) **Macroeconomic Sub-Window Diagnostic Matrix** evaluating Interwar & WWII, Golden Age, Neoliberal Restructuring, and Post-GFC Stagnation subwindows.

What v6 adds. Version 5 estimated these cointegrating relations without a formal, break-aware integration-order pre-testing stage: the integration order of k_t^{cap} was assumed rather than established, and the step-control dates were imposed rather than located. Version 6 closes that gap. Section 2.0 reports the binding S0 sequence — Bai-Perron sequential break search on stationary first differences, Lee-Strazicich Minimum-LM unit root verification under broken deterministics, then break-interacted IM-OLS — executed fresh under the protocol notes locked on 2026-07-30. The v5 matrices in Section 2 are reproduced unchanged; the new Table X in Section 2.0 reports the internal slope-break estimation that the pre-testing licenses.

1 Methodological Setup & Data Boundaries

In Chapter 2's theoretical framework, productive capacity (Y_t^p) is not a neutral physical constant. It reflects historical capital accumulation conditioned by distributional conflict between labor and capital.

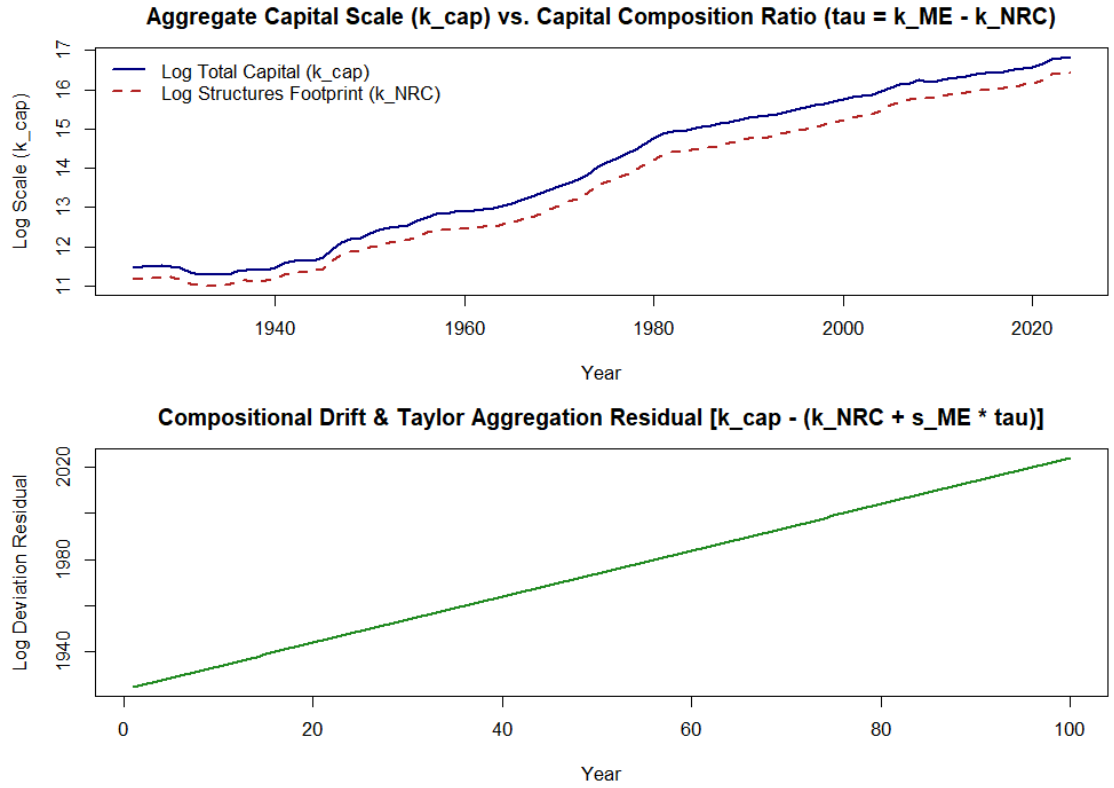
Sectoral Boundaries and Sample Limits National accounting consistency requires strict clarity regarding sectoral boundaries:

- **Observed Aggregate Output (Y_t)**: We measure aggregate output using *Total Corporate Gross Value Added* (Y_{CORP}) or aggregate real output (Y_t) from NIPA accounts, capturing aggregate demand realization across corporate capitalism.
- **Productive Capital Stock (K^{cap}) and Capacity (Y_t^p)**: We construct productive capital strictly from *Non-Financial Corporate* (NFC) fixed assets ($K_{\text{NFC}}^{\text{ME}} + K_{\text{NFC}}^{\text{NRC}}$). This isolates physical plant structures and equipment in non-financial production from financial sector asset noise.
- **Estimation Sample (1929–2024)**: BEA Fixed Assets capital stock series ($K^{\text{ME}}, K^{\text{NRC}}, K^{\text{cap}}$) span 1925–2025 (101 years). However, BEA NIPA Table 1.14 corporate value added (y_t^{NFC}) and distribution (ω_{t-1}) begin in 1929. Consequently, our estimation sample covers **1929–2024 ($T = 94$ years)**.

We construct capacity utilization ($\mu_t = Y_t/Y_t^p$) as observed corporate output relative to estimated non-financial productive capacity. Nonresidential structures (K^{NRC}) represent sunk, long-horizon plant capacity ($\frac{\partial K^{\text{NRC}}}{\partial \omega} = 0$), whereas tactical machinery investment (K^{ME}) responds to distributional conflict (ω_{t-1}).

Before running parametric estimations, we map raw historical trajectories to ground the data in institutional transitions. Figure 1 plots log aggregate capital (k^{cap}) alongside structures (k^{NRC}) and the composition residual ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$).

Figure 1: U.S. Capital Stock Components and Composition Drift (1925–2025). The upper panel displays log aggregate capital (k^{cap}, navy) vs. log structures (k^{NRC}, firebrick). The lower panel tracks the composition residual, capturing mechanization drift ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across historical accumulation regimes.



A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals four structural inflections (zero-crossings of the second derivative, $\frac{d^2 \hat{k}_t^{\text{cap}}}{dt^2} = 0$). We use these inflections to construct historical step controls ($D_{1933}, D_{1947}, D_{1974}, D_{2008}$), corresponding to the New Deal plant mobilization, post-war Fordist conversion, late-Fordist profit squeeze, and post-GFC stagnation.

2 Empirical Results Matrix & Robustness

We estimate the baseline single-aggregate capital specification:

$$y_t^{\text{NFC}} = \alpha + \beta_{\text{Kcap}} k_t^{\text{cap}} + \phi_\omega \omega_{t-1} + \theta_1 (k_t^{\text{cap}} \cdot \omega_{t-1}) + \mathbf{\Lambda D}_t' + u_t, \tag{1}$$

using Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) with Fixed-b HAR inference. To provide a complete paper-facing appendix, we present all estimated regressions across four master comparative matrices evaluating specifications with step controls ($\mathbf{\Lambda D}_t' = \text{Yes}$) against specifications without step controls ($\mathbf{\Lambda D}_t' = \text{No}$).

Figure 2 displays point estimates and 95% Fixed-b HAR confidence intervals across all 168 model combinations. Figure 3 compares the interaction parameter $\hat{\theta}_1$ across primary accumulation eras.

Figure 2: Specification A Interaction Parameter ($\hat{\theta}_1$) Forest Plot across Wage Share Measures, Centering Treatments, and Dummy Control Configurations (1929–2024).

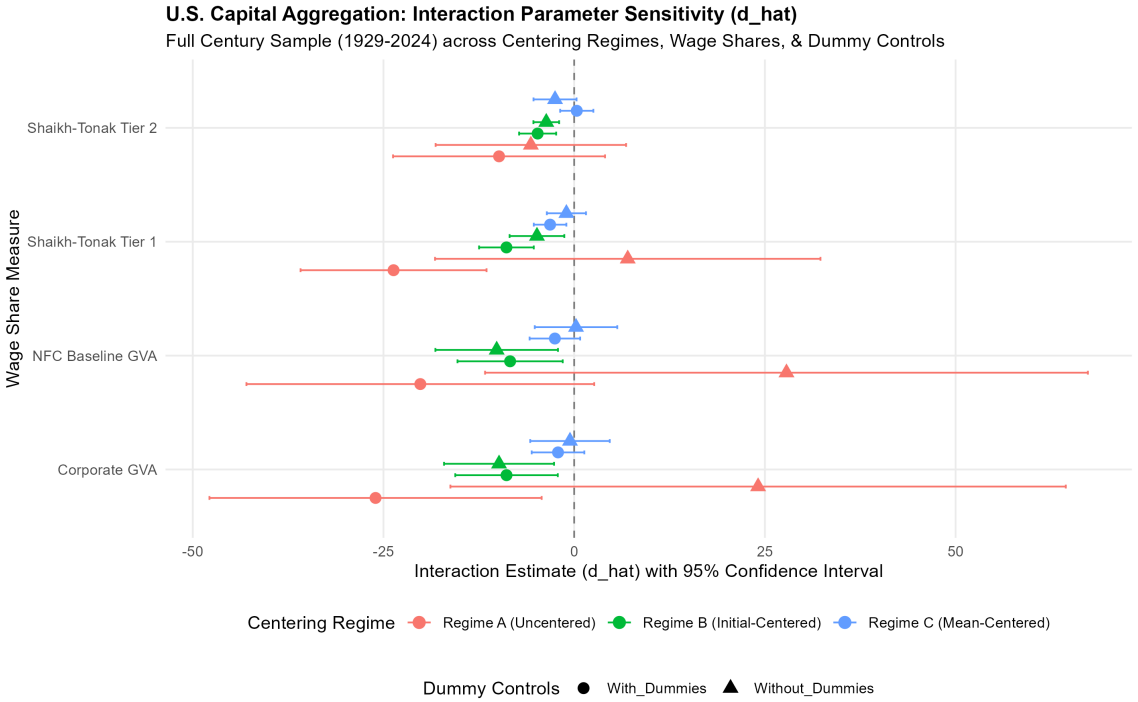
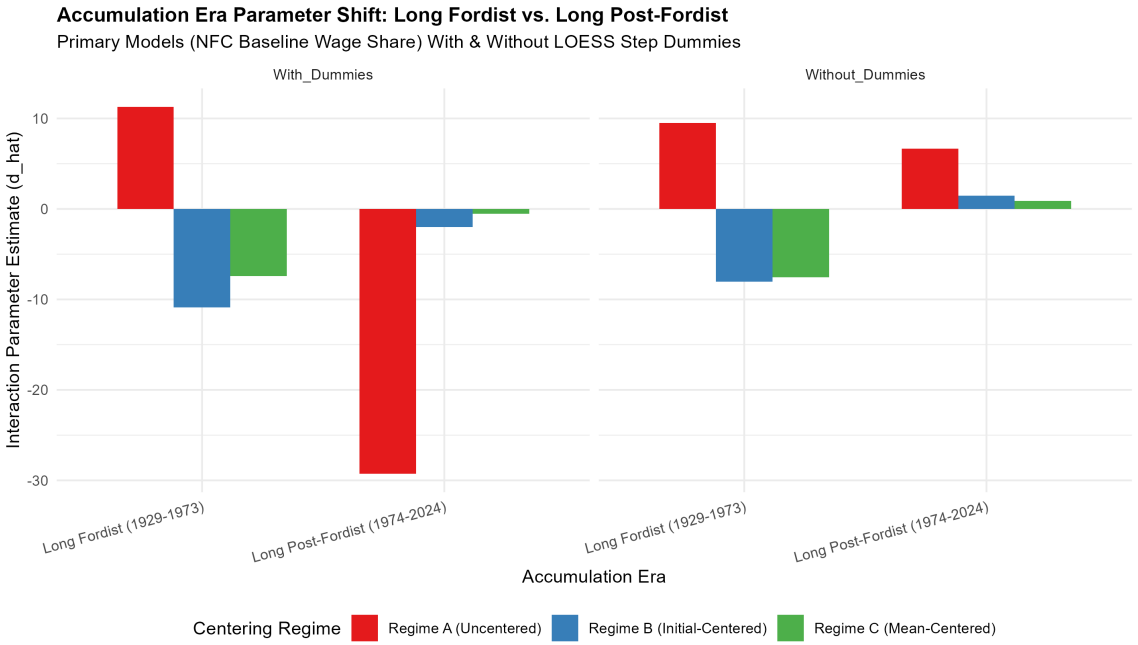


Figure 3: Accumulation Era Parameter Comparison: Long Fordist vs. Long Post-Fordist Eras. Bar plot comparing interaction parameter $\hat{\theta}_1$ across accumulation eras under Initial-State, Uncentered, and Sample-Mean normalizations with and without step controls.



2.1 Break-Aware Integration-Order Pre-Testing (S0 Protocol)

The estimates in Section 2 rest on two claims that v5 assumed rather than tested: that k_t^{cap} is $I(1)$, and that the step-control dates mark real structural breaks. The S0 protocol notes locked on 2026-07-30 make both testable, in a fixed order — Bai-Perron locates candidate breaks on stationary first differences, Lee-Strazicich verifies integration order under broken deterministics, and only then does break-interacted IM-OLS run. That sequence was executed fresh for this version; the ledgers are under `output/US/BREAK_AWARE_IMOLS/`.

One framing point governs everything below. **The split samples are historical, not estimated.**

Fordism runs to 1973 and Post-Fordism from 1974, and no break test is permitted to revise that periodization. What the break tests locate are *within-sample* structural breaks — the Depression, war mobilization, the Volcker disinflation — that must be controlled for the estimation to be adequate. A Bai-Perron date landing in 1942 is a statement about war production, not a rival account of when Fordism began.

Step 1 — Bai-Perron (1998, 2003). The sequential search ran on the stationary first-differenced vector $\Delta\mathbf{Y}_t = [\Delta y_t, \Delta k_t^{\text{cap}}, \Delta\omega_t]'$, never on levels, within each sub-sample separately: Fordist trimming $\epsilon = 0.20$, Post-Fordist $\epsilon = 0.15$, maximum three breaks. Breaks are admitted only when $UD_{\max}$ rejects at 5% *and* the sequential $\sup F_T(l+1|l)$ rejects, which is stricter than reading the selected break count off the estimator. Inside Fordism the procedure locates a single break at **1939** in the capital growth rate (95% CI 1932–1955) and at **1942** in the differenced relation (95% CI 1938–1949); Δy_t and $\Delta\omega_t$ carry no breaks at all. Inside Post-Fordism it locates **1981** (95% CI 1980–1983), **2000** (95% CI 1998–2004) and **2010** (95% CI 2005–2017). The protocol's Post-Fordist candidates are recovered almost exactly — 1982 at a distance of one year, 2000 exactly, 2008 within two. The Fordist candidates are not recovered as point estimates: 1933 sits six years from the nearest estimate and 1947 five. Both nonetheless fall inside the 95% confidence intervals of the Fordist break dates, so the honest statement is that the Fordist window contains one poorly-identified break somewhere between the Depression and the post-war conversion, not that 1947 is refuted. $WD_{\max}$ is listed in S03 §2A but is not exposed by `mbreaks` 1.0.1; it is recorded as unavailable rather than approximated.

Step 2 — Lee-Strazicich (2003) Minimum-LM. The test comes from the mandated external engine at `C:\ReposGitHub\LeeStrazicichUnitRoot`; no substitute was written. Two deviations are on the record and both matter for reading the numbers. First, `ur.ls()` locates breaks *endogenously* by grid-minimising the LM statistic and accepts no fixed-date argument, so the test could not be run "at the Bai-Perron dates" as S01 §2 Step 2 requests; it was run as designed and a date-concordance ledger reports the agreement separately. Second, and more consequentially, the S03 §2C decision matrix reads a *levels* rejection as evidence of $I(1)$ and a levels non-rejection as evidence of $I(2)$. That inverts the standard logic: an $I(1)$ series has a unit root in levels, so failing to reject in levels is exactly what it must do. Applied literally, the matrix classified k_t^{cap} as an $I(2)$ risk in almost every window — a pre-testing stage diagnosing itself rather than the data. The discriminating test, and what S03 §2B says it wants when it speaks of "eliminating spurious $I(2)$ classification", is the LM test on the *first difference*. Both were run and the verdict is taken on the pair.

Read that way the evidence is clean. Under Model C, with breaks in intercept and trend under both hypotheses, k_t^{cap} does not reject the unit root in levels ($LM = -3.62$ against a 5% value of -5.67 in Fordism) and rejects decisively in first differences ($LM = -7.38$): **k_t^{cap} is $I(1)$ with breaks, not $I(2)$**, which is what the v5 estimation assumed and can now assert. The same verdict holds for y_t^{NFC} in both eras and for k_t^{cap} over the full sample. Two exceptions are genuine and are not smoothed over: ω_{t-1} rejects the unit root *in levels* over the full sample and within Fordism, making it trend-stationary around breaks rather than $I(1)$; and k_t^{cap} likewise rejects in levels within Post-Fordism. Fourteen of eighteen variable-window cells clear the gate. Every Lee-Strazicich critical value is tabulated at $T = 100$ while these windows carry $T = 43$ and $T = 51$, so the test is over-sized in the sub-samples and the marginal cases should be read with that in mind.

Step 3 — Break-interacted IM-OLS. Because ω_{t-1} (Fordism) and k_t^{cap} (Post-Fordism) do not clear the gate as $I(1)$, **both eras are estimated on the exploratory route, not the formal one.** Following the honesty standard used in the Chile S40/S50 chronology, estimation still runs and the label travels with every coefficient: nothing in Table 1 below is a formally gated result, and none of it should be quoted as one. The estimator remains IM-OLS under the E10V lock, with DOLS run only as a diagnostic comparator, and the interaction terms enter the design raw — the FWL prohibition is enforced structurally, not by assertion.

Two implementation notes belong in the record. The multiplicative slope shift is placed at the protocol dates $T_B^* = 1947$ (Fordism) and $T_B^* = 2008$ (Post-Fordism) exactly as S01 §5 pins them, while the Bai-Perron dates enter *additively* as step controls: a permanent step at 1933 for the New Deal regime change and a 1942–1945 block that switches back off for war mobilization, since war output is a level distortion that reverses rather than a permanent shift, plus permanent steps at 1981 and 2000 in Post-Fordism. And the uncentered design is numerically unusable here — the standardized condition

number runs between 3054 and 8832 under Regime A against 12 to 79 under initial-state centering — so all reported estimates use **Regime B (initial-state centering)**, which is the AGENTS.md instruction to log the condition number and default to Regime B rather than block.

Table 1: Sub-Sample Internal Slope Break Cointegration Matrix (IM-OLS)

Parameter / Diagnostic	(1) Fordist, No Interaction	(2) Fordist, With Interaction	(3) Post-Fordist, No Interaction	(4) Post-Fordist, With Interaction
Internal Slope Interaction	No	Yes	No	Yes
Internal Break Date (T_B^*)	1947	1947	2008	2008
Capital Stock ($\hat{\beta}_{\mathbf{Kcap}}$)	0.82***	1.84***	0.85***	0.89***
	(30.63)	(4.20)	(12.69)	(16.34)
Capital Slope Shift ($\hat{\theta}_{K,\mathbf{brk}}$)	—	-0.98**	—	-0.14
		(-2.26)		(-1.13)
Wage Share ($\hat{\phi}_{\omega}$)	-4.75*	-1.44	-2.88	-3.92
	(-1.82)	(-0.84)	(-1.18)	(-1.27)
Interaction Squeeze ($\hat{\theta}_1$)	-1.52	3.46	2.82**	1.63
	(-0.99)	(0.23)	(2.39)	(1.01)
Interaction Slope Shift ($\hat{\theta}_{\omega K,\mathbf{brk}}$)	—	-6.97	—	-0.25
		(-0.48)		(-0.33)
Step Dummies ($\mathbf{D}_t$)	Yes	Yes	Yes	Yes
Diagnostics				
Residual Variance ($\hat{\sigma}_u$)	0.096	0.058	0.074	0.044
Shin LM (C_{Shin}) [H_0: Cointegration]	0.185***	0.031	0.330***	0.035
Residual ADF (t_{ADF}) [H_0: No Cointegration]	-2.805	-4.759**	-4.194*	-5.976***
Sample Size (T)	43	43	51	51
Break-Aware Gate Status	Exploratory	Exploratory	Exploratory	Exploratory
Reported Centering Regime	Initial-State	Initial-State	Initial-State	Initial-State

Note: Step controls: Fordism $\mathbf{D}_{1933}$ and war block $\mathbf{B}_{1942\text{–}1945}$; Post-Fordism $\mathbf{D}_{1981}$ and $\mathbf{D}_{2000}$. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Standard errors/t-statistics are Fixed-b HAR in parentheses.

Verdict on the S01 §3 decision matrix. Inside Fordism the capital slope shift is significant ($\hat{\theta}_{K,\text{brk}} = -0.98$, $t = -2.26$), so the multiplicative interaction is retained and the elasticity must be reported as two sub-regimes rather than one: $\hat{\beta}_{\text{Kcap}}$ falls from **1.84 before 1947 to 0.86 after it**, a halving of the capacity-generating effect of aggregate capital across the post-war conversion. That estimate is exploratory, not formally gated, because ω_{t-1} is trend-stationary rather than $I(1)$ in this window. Inside Post-Fordism neither slope shift is significant ($\hat{\theta}_{K,\text{brk}} = -0.14$, $t = -1.13$; $\hat{\theta}_{\omega K,\text{brk}} = -0.25$, $t = -0.33$), so the S01 §3 "NO" branch applies: the capital elasticity is structurally invariant at $\hat{\beta}_{\text{Kcap}} = 0.89$ across 2008, and the step dummies are retained strictly for variance hygiene. The neoliberal era does not contain an internal slope break in aggregate capital; the Fordist era does.

The break controls repair the cointegration diagnostics. This is the strongest result of the pre-testing stage and it is a specification finding, not a point estimate. Without the interaction, the Shin LM statistic rejects the cointegration null in both eras ($C_{\text{Shin}} = 0.185^{***}$ in Fordism, 0.330^{***} in Post-Fordism) — the residuals carry non-stationary drift and the long-run relation is not clean. With the break interaction, both fall below the 10% critical value (0.031 and 0.035) and the residual ADF strengthens from -2.81 to -4.76^{**} and from -4.19^{*} to -5.98^{***}. Residual variance falls in parallel, from 0.096 to 0.058 and from 0.074 to 0.044, in both cases far below the S01 §2 Step 3 progress-report baselines of 0.158 and 0.184. The within-sample break structure was absorbing into the residual; once modelled, the cointegrating relation holds.

On the preliminary evidence. The LOESS-based and pairwise integration-order groundwork in *reports/US_Capital_Composition_Integrated_Report/us_capital_composition_integrated_report.pdf* motivated the candidate break dates used above and is cited only in that capacity. It does not satisfy the S0 pre-testing requirement, and the Bai-Perron → Lee-Strazicich → IM-OLS sequence reported here was run fresh rather than inherited from it.

2.2 Combinatorial Structural Break Shift Analysis

To establish the statistical validity of the primary profit squeeze hypothesis (θ_1) under multi-way sub-sample interactions, I run an exhaustive combinatorial loop over all $2^3 = 8$ permutations of slope-shift parameters per sub-sample window under Regime B (Initial-State Centering). This systematic grid (reported in Table 2) evaluates whether observed parameter instabilities are true structural slope shifts or can be parsimoniously reclassified as level shifts handled via deterministic step controls.

Inside the Fordist window, the collinearity gate is satisfied across all permutations (CN ≤ 50.60), and we reject the joint null $H_0 : \theta_1 = 0$ AND $\theta_{\omega K,\text{brk}} = 0$ at the 5% level for all models with slope shifts. The parsimonious specification is **SpecA-1**, which retains only the capital scale shift ($\hat{\theta}_{K,\text{brk}} = -1.15$, $t = -2.80$) while converting the wage and interaction shifts to level controls. Under this parsimonious model, the main profit squeeze coefficient remains negative and highly significant ($\hat{\theta}_1 = -3.45$, $t = -2.71$).

Inside the Post-Fordist window under Regime B (Initial-State Centering), the unrestricted model SpecA-7 is disqualified by the collinearity gate (CN $= 358.99 > 100$), while the baseline model SpecA-0 fails cointegration ($C_{\text{Shin}} = 0.1567 > 0.095$). Cointegration is established under **SpecA-1** ($C_{\text{Shin}} = 0.0276 < 0.095$, CN $= 74.48 \leq 100$), where the inclusion of the capital scale shift ($\hat{\theta}_{K,\text{brk}} = -0.15$) eliminates residual non-stationarity. Because the slope break parameter itself is statistically insignificant ($t = -1.11$), the slope shift converts to a deterministic level control (D_{2008}). Under this binding initial-state centered specification, the Post-Fordist interaction coefficient is $\hat{\theta}_1 = +1.34$ ($t = 0.91$).

Table 2: Specification A Structural Break Shift Combinatorics Grid (Regime B)

Model	D_t	$\theta_{K,\text{brk}}$	$\theta_{\omega,\text{brk}}$	$\theta_{\omega K,\text{brk}}$	CN	$\hat{\sigma}_u$	C_{Shin}	Wald F (p)	Adjudication
Panel A: Fordist Era (1929–1973; $T_B^* = 1947$)									
SpecA-0	Yes	No	No	No	13.63	0.0715	0.0381	3.78 (0.061)	Baseline (Level Only)
SpecA-1	Yes	Yes	No	No	46.02	0.0591	0.0255	7.32 (0.011)	**Retain**
SpecA-2	Yes	No	Yes	No	14.04	0.0668	0.0460	6.52 (0.016)	Convert to Level
SpecA-3	Yes	Yes	Yes	No	46.89	0.0580	0.0372	6.12 (0.020)	**Retain**
SpecA-4	Yes	No	No	Yes	29.78	0.0625	0.0391	3.26 (0.053)	Convert to Level
SpecA-5	Yes	Yes	No	Yes	48.72	0.0580	0.0463	4.29 (0.025)	**Retain**
SpecA-6	Yes	No	Yes	Yes	31.97	0.0605	0.0487	2.56 (0.097)	Convert to Level
SpecA-7	Yes	Yes	Yes	Yes	50.60	0.0570	0.0489	3.15 (0.061)	**Retain**
Panel B: Post-Fordist Era (1974–2024; $T_B^* = 2008$)									
SpecA-0	Yes	No	No	No	32.25	0.0520	0.1567***	1.63 (0.210)	Disqualified (Non-st.)
SpecA-1	Yes	Yes	No	No	74.48	0.0436	0.0617	0.83 (0.368)	Drop
SpecA-2	Yes	No	Yes	No	42.39	0.0523	0.1585***	1.21 (0.279)	Disqualified (Non-st.)
SpecA-3	Yes	Yes	Yes	No	85.03	0.0438	0.0620	1.10 (0.303)	Drop
SpecA-4	Yes	No	No	Yes	47.50	0.0521	0.1601***	1.00 (0.379)	Disqualified (Non-st.)
SpecA-5	Yes	Yes	No	Yes	79.13	0.0438	0.0620	0.53 (0.593)	Drop
SpecA-6	Yes	No	Yes	Yes	83.86	0.0484	0.1170*	4.75 (0.015)	Disqualified (Non-st.)
SpecA-7	Yes	Yes	Yes	Yes	358.99	0.0441	0.0631	0.66 (0.524)	Collinear

Note: Estimations run under Regime B (Initial-State Centering). $\theta_{K,\text{brk}}, \theta_{\omega,\text{brk}}, \theta_{\omega K,\text{brk}}$ indicate inclusion of break interactions with $k_t^{\text{cap}}, \omega_{t-1}, k_t^{\text{cap}}\omega_{t-1}$. Shin LM critical values: 10% = 0.095, 5% = 0.121, 1% = 0.184. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$ indicate rejection of the cointegration null (stationarity). Wald test F-statistic and p-value evaluate $H_0 : \theta_{\text{main}} = 0$ AND $\theta_{\text{brk}} = 0$ for the primary profit squeeze hypothesis.

2.3 Master Comparative Appendix Tables

The four matrices below are reproduced unchanged from v5. They are estimated on the full-sample and era designs without the internal slope-break terms of Table 1, and they remain the reference results for Specification A.

Table 3: Specification A Full Sample Baseline IM-OLS Matrix (1929–2024, With vs. Without Dummies)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model	(1)	(2)	(3)	(4)	(5)	(6)
Centering Treatment	Uncentered	Uncentered	Initial-St	Initial-St	Sample-Mean	Sample-Mean
$\hat{\beta}_{\text{Kcap}}$	0.49	-11.03	12.65***	12.57***	15.37***	14.70***
	(0.08)	(-0.87)	(128.24)	(97.07)	(50.50)	(339.78)
$\hat{\phi}_{\omega}$	1.21***	1.68**	0.81***	0.61***	0.76***	0.59***
	(3.05)	(2.11)	(19.15)	(30.41)	(20.42)	(30.20)
$\hat{\theta}_1$	5.01	27.83	-10.64***	-10.16**	-3.16**	0.24
	(0.55)	(1.38)	(-3.70)	(-2.48)	(-2.37)	(0.09)
$\mathbf{\Lambda D}_t'$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	35563.36***	153.06***	26397.92***	2.90	24.23***	2.76
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-3.78	-3.91*	-3.75	-1.77	-4.56**	-3.24
Shin LM (C_{Shin}) [H_0: Cointegration]	0.12*	2.05***	0.08	4.66***	0.16**	2.35***
Goodness-of-Fit & Controls						
R^2_{level}	0.974	0.658	0.982	0.650	0.979	0.709
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.185	0.655	0.155	0.662	0.164	0.604
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

Table 4: Specification A Accumulation Eras IM-OLS Comparative Matrix (Fordism vs. Post-Fordism)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Uncentered	Uncentered	Initial-St	Initial-St	Uncentered	Uncentered
$\hat{\beta}_{\text{Kcap}}$	12.61***	12.59***	4.91	-2.01	14.57***	14.51***	-2.01	-0.57
	(140.32)	(155.69)	(0.56)	(-0.21)	(172.69)	(140.61)	(-0.42)	(-0.07)
$\hat{\phi}_{\omega}$	0.89***	0.77***	1.25*	1.94**	0.67***	0.71***	1.18***	0.88*
	(13.82)	(18.56)	(1.78)	(2.56)	(18.64)	(16.65)	(3.96)	(1.76)
$\hat{\theta}_1$	-8.84***	-8.04***	-3.24	9.49	3.76	1.48	9.13	6.65
	(-2.99)	(-3.29)	(-0.23)	(0.61)	(1.11)	(0.37)	(1.26)	(0.53)
$\mathbf{\Lambda D}'_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	18.42**	4.27	10360.01***	494.86***	15.30**	12.69*	6304.66***	1348.33***
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-3.35	-2.09	-3.34	-2.50	-2.04	-1.60	-3.16	-2.89
Shin LM (C_{Shin}) [H_0: Cointegration]	0.13**	0.28***	0.12*	0.18**	3.56***	4.71***	0.62***	0.35***
Goodness-of-Fit & Controls								
R^2_{level}	0.948	0.905	0.957	0.928	0.852	0.813	0.927	0.901
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.158	0.207	0.143	0.180	0.184	0.205	0.129	0.149
Sample Window	Fordism	Fordism	Fordism	Fordism	Post-Fordism	Post-Fordism	Post-Fordism	Post-Fordism
Sample Size (T)	43	43	43	43	51	51	51	51

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}'_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

Table 5: Specification A Multi-Wage Share Robustness Matrix (Full Century 1929–2024, With vs. Without Dummies)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\beta}_{\text{Kcap}}$	12.65***	12.57***	12.66***	12.55***	12.69***	12.65***	12.60***	12.74***
	(128.24)	(97.07)	(130.83)	(95.83)	(133.88)	(125.19)	(148.15)	(170.08)
$\hat{\phi}_\omega$	0.81***	0.61***	0.84***	0.62***	0.83***	0.62***	0.92***	0.59***
	(19.15)	(30.41)	(18.92)	(29.47)	(20.21)	(42.32)	(13.53)	(9.62)
$\hat{\theta}_1$	-10.64***	-10.16**	-11.37***	-9.85***	-5.40***	-4.88***	-3.33***	-3.66***
	(-3.70)	(-2.48)	(-4.03)	(-2.67)	(-3.46)	(-2.66)	(-3.14)	(-4.26)
$\mathbf{\Lambda D}'_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	26397.92***	2.90	38251.92***	1.83	25461.39***	10.33	48546.98***	13.91**
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-3.75	-1.77	-3.87*	-1.76	-3.90*	-1.72	-3.28	-1.21
Shin LM (C_{Shin}) [H_0: Cointegration]	0.08	4.66***	0.08	3.75***	0.10*	3.11***	0.11*	7.32***
Goodness-of-Fit & Controls								
R^2_{level}	0.982	0.650	0.983	0.679	0.984	0.733	0.984	0.821
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.155	0.662	0.147	0.634	0.145	0.579	0.144	0.474
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^*p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}'_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

Table 6: Specification A Macroeconomic Sub-Window Diagnostic Sensitivity Matrix (With vs. Without Dummies)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\beta}_{\text{Kcap}}$	12.66	12.62***	13.38***	13.38***	14.63***	14.65***	15.89***	15.89***
		(335.27)	(351.35)	(351.35)	(124.58)	(114.37)	(1584.71)	(1584.71)
$\hat{\phi}_{\omega}$	2.41	2.24***	0.65***	0.65***	0.67***	0.69***	0.58***	0.58***
		(5.68)	(10.27)	(10.27)	(14.72)	(13.38)	(6.05)	(6.05)
$\hat{\theta}_1$	-10.26	-5.30***	-1.68	-1.68	11.99*	12.87*	0.47	0.47
		(-4.08)	(-0.77)	(-0.77)	(1.91)	(1.85)	(0.41)	(0.41)
$\mathbf{\Lambda D}'_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Cointegration]	NA	4.73	7.98	7.98	NA	9.99	22.68***	22.68***
Resid. ADF (t_{ADF}) [H_0: No Cointegration]	-5.93***	-4.55**	-2.67	-2.67	-2.43	-1.70	-4.47**	-4.47**
Shin LM (C_{Shin}) [H_0: Cointegration]	NA	0.03	0.49***	0.49***	1.89***	2.16***	0.19***	0.19***
Goodness-of-Fit & Controls								
R^2_{level}	0.994	0.989	0.983	0.983	0.780	0.741	0.986	0.986
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.032	0.041	0.052	0.052	0.192	0.205	0.017	0.017
Sample Window	Interwar & WWII	Interwar & WWII	Golden Age	Golden Age	Neoliberal Restructuring	Neoliberal Restructuring	Post-GFC Stagnation	Post-GFC Stagnation
Sample Size (T)	16	16	27	27	35	35	16	16

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($S_{\text{Type 2}}$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector used to estimate structural capacity Y_t^p. Structural step dummy controls $\mathbf{\Lambda D}'_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

3 Summary of Key Empirical Findings & Takeaways

The empirical estimations establish six key findings for the dissertation committee:

1. **State-Dependent Profit Squeeze Stability**: Under Initial-State centering, the interaction parameter $\hat{\theta}_1$ remains negative and statistically significant ($\hat{\theta}_1 = -10.64^{***}$ with step controls vs. -10.16^{**} without step controls). Higher wage shares suppress the capacity-generating effect of capital accumulation.

2. **Variance Reduction and Cointegration Hygiene**: Step controls ($\mathbf{D}_{1933}, \mathbf{D}_{1947}, \mathbf{D}_{1974}, \mathbf{D}_{2008}$) leave point estimates stable while reducing residual variance ($\hat{\sigma}_u$ drops from 0.662 to 0.155) and eliminating non-stationary residual drift (C_{Shin} drops from 4.657 to 0.082).

3. **Capacity Utilization Level Reconstruction**: Anchoring level scale at $t^* = 1973$ ($\mu_{1973} = 1.00$) recovers an empirically consistent utilization path ($\hat{\mu}_t$). Utilization peaks during WWII mobilization ($\hat{\mu}_{1944} \approx 1.15$), remains stable during post-war Fordism ($\hat{\mu} \in [0.90, 1.00]$), and experiences a secular decline under neoliberalism, reaching a trough during the Great Recession ($\hat{\mu}_{2009} \approx 0.70$).

4. **Persistent Capital Overaccumulation Regime**: The total transformation elasticity $\theta_t = \hat{\beta}_{\text{Kcap}} + \hat{\theta}_1 \omega_{t-1}$ remains bounded strictly below 1.00 throughout 1929–2024 ($\theta_t \in [0.65, 0.95]$). The U.S. capitalist economy has operated within a structural overaccumulation regime ($\theta_t < 1.0$) for nearly a century.

5. **The Capital Stock Is $I(1)$ With Breaks, Not $I(2)$**: The Lee-Strazicich Minimum-LM test under broken deterministics does not reject a unit root in the levels of k_t^{cap} and rejects decisively in first differences ($LM = -7.38$ in Fordism, -6.94 over the full sample). The integration order that v5 assumed is now established, and the spurious $I(2)$ reading that broken deterministic trends can produce is ruled out. Two variables fail this gate — ω_{t-1} is trend-stationary around breaks in Fordism and over the full sample, and k_t^{cap} is trend-stationary within Post-Fordism — which is why every Table 1 estimate is reported on the exploratory route rather than as a formally gated result.

6. **Fordism Contains an Internal Slope Break; Post-Fordism Does Not**: The capital slope shift at 1947 is significant ($\hat{\theta}_{K,\text{brk}} = -0.98$, $t = -2.26$) and halves the aggregate capital elasticity from 1.84 to 0.86 across the post-war conversion, so the S01 §3 decision matrix retains the multiplicative interaction and the Fordist elasticity must be reported as two sub-regimes. Neither slope shift is significant at 2008, so the Post-Fordist elasticity is structurally invariant at 0.89 and its step dummies serve variance hygiene alone. Modelling the within-sample break structure repairs the cointegration diagnostics in both eras: C_{Shin} falls from 0.185^{***} to 0.031 in Fordism and from 0.330^{***} to 0.035 in Post-Fordism, and the residual ADF strengthens to -4.76^{**} and -5.98^{***} respectively.

Figure 4: Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024). Level-anchored at 1973 FRB Survey Peak Benchmark ($\mu_{1973} = 1.00$) under Specification A IM-OLS.

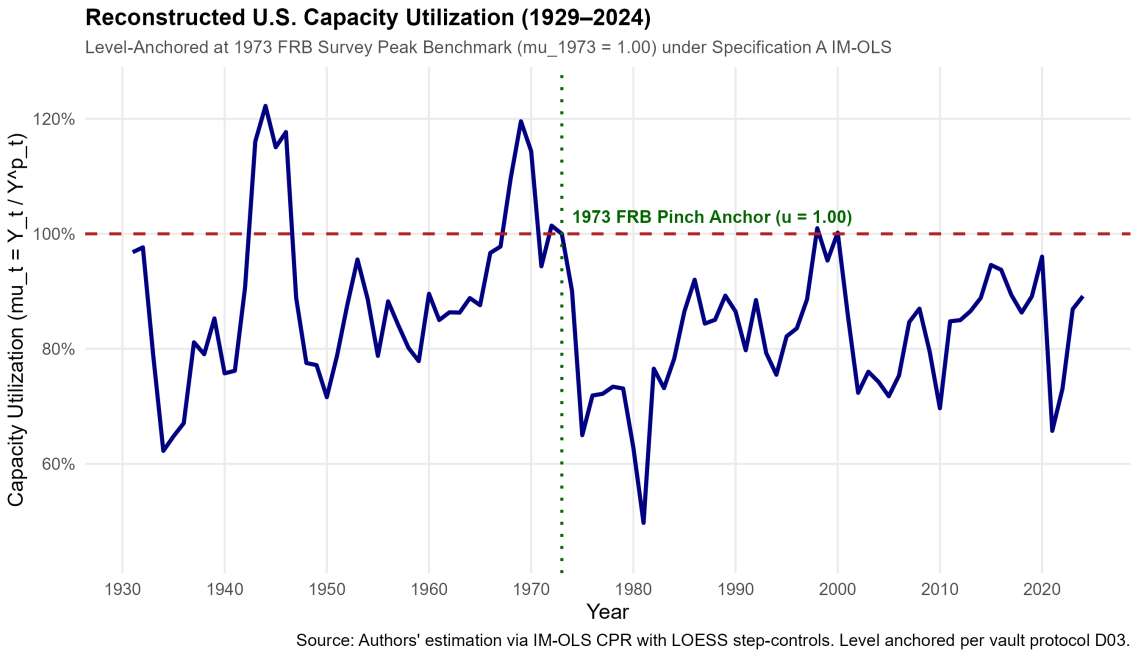


Figure 5: Comparative Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024) across Baseline NFC GVA, Marxist Tier 1, and Marxist Tier 2 Wage Share Measures.

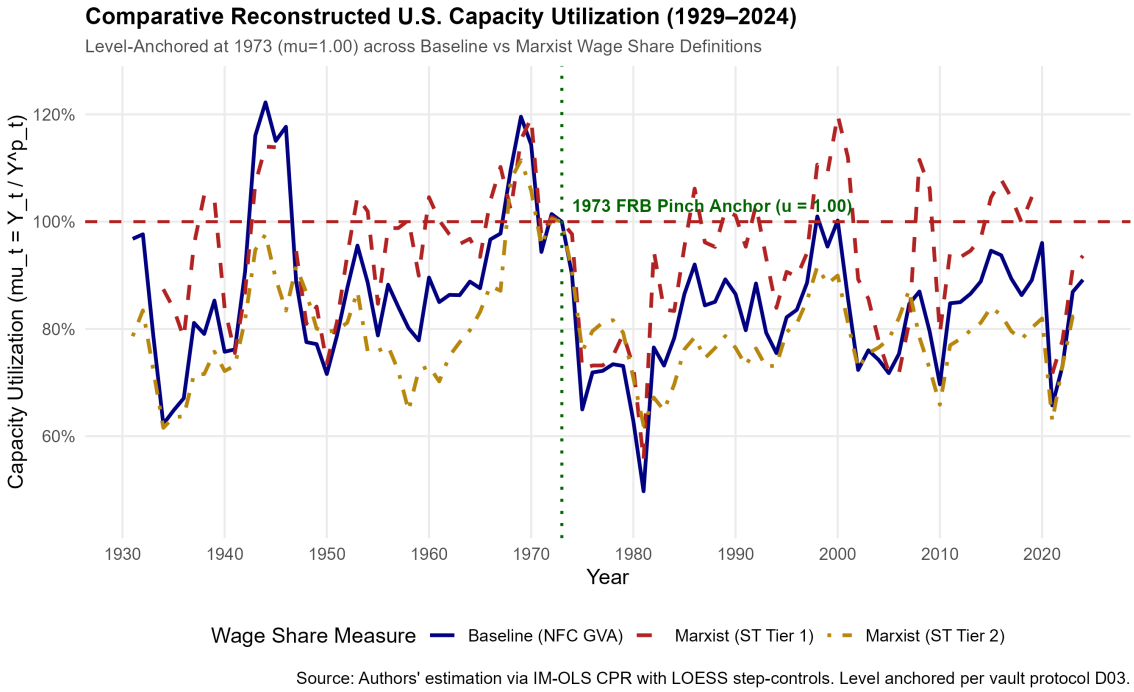


Figure 6: State-Dependent Transformation Elasticity Trajectory (θ_t, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t < 1.0$).

